TEXORA · LECTURE SHEET

Dimensional Analysis and Similitude

How to learn from models: every physical equation must balance dimensionally, and Buckingham's Pi theorem turns a shopping list of variables into a few dimensionless groups — Reynolds, Froude, drag coefficient — that let a small model speak truthfully about a large prototype.

ME-3210 · SOLID MECHANICS AND MACHINE DESIGN · TOPIC 4 · LEVEL 3 TERM 2

1. Dimensions and homogeneity

Every mechanical quantity is built from three fundamental dimensions — mass M, length L, time T (temperature Θ joins for thermal problems): velocity LT^{-1}, force MLT^{-2}, pressure $ML^{-1}T^{-2}$, viscosity $ML^{-1}T^{-1}$, power $ML^{2}T^{-3}$. A physically meaningful equation must be **dimensionally homogeneous**: both sides, and every added term, carry identical dimensions. Checking homogeneity is the engineer's spell-checker — it catches inverted fractions and missing densities before they cost money — and it is the foothold for the powerful method that follows.

2. Buckingham's Pi theorem

Theorem: if a problem involves n variables formed from m fundamental dimensions, the physics can be written as a relation among (n − m) independent **dimensionless groups** (Π groups). Recipe: list the variables; choose m **repeating variables** that among themselves contain all the dimensions and cannot form a dimensionless group alone; combine each remaining variable with the repeating set to force dimensionlessness; interpret the groups.

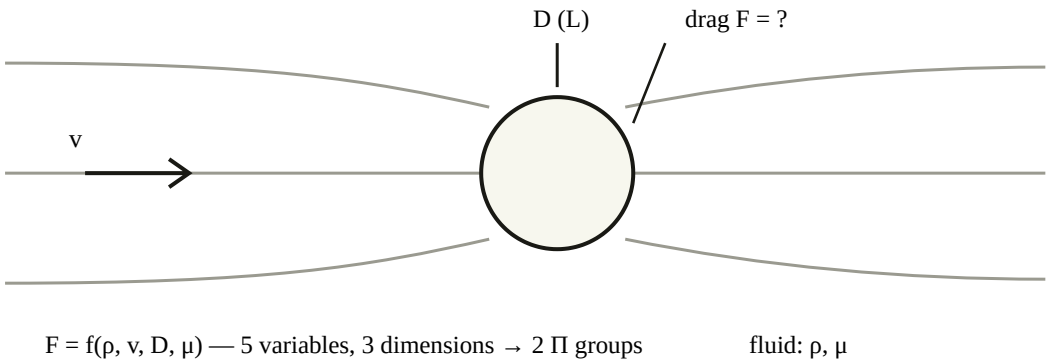


Figure 1 — Flow past a sphere: the drag problem that the worked derivation reduces to two dimensionless groups.

WORKED DERIVATION — DRAG ON A SPHERE

Variables: F [MLT^{-2}], ρ [ML^{-3}], v [LT^{-1}], D [L], μ [$ML^{-1}T^{-1}$]. n = 5, m = 3 → **two Π groups**.
Repeating variables: ρ, v, D.
Π_1 from F: F ρ^a v^b D^c dimensionless → M: 1 + a = 0 → a = −1; T: −2 − b = 0 → b = −2; L: 1 − 3a + b + c = 1 + 3 − 2 + c = 0 → c = −2. Π_1 = **$F/(\rho v^2 D^2)$** — the drag-coefficient family.
Π_2 from μ: μ ρ^a v^b D^c → a = −1 (M), b = −1 (T: −1 − b = 0); L: −1 − 3a + b + c = −1 + 3 − 1 + c = 0 → c = −1. $\Pi_2 = \mu/(\rho v D)$ = **1/Re**.
Conclusion: $F/(\rho v^2 D^2) = \varphi(Re)$ — one curve of drag coefficient against Reynolds number replaces an infinity of experiments over five variables. This is exactly how sphere, cylinder and aerofoil data are published, and how Topic 3's friction factors were organised on the Moody chart.

3. The famous groups

GROUP	DEFINITION	PHYSICAL MEANING / WHERE IT RULES
Reynolds Re	$\rho v L/\mu$	Inertia vs viscosity — pipe regime, drag, boundary layers
Froude Fr	$v/\sqrt{(gL)}$	Inertia vs gravity — free-surface flow, ships, spillways, open channels
Euler Eu	$\Delta p/(\rho v^2)$	Pressure vs inertia — pressure-drop coefficients, cavitation number kin
Mach M	v/a	Compressibility — gas flow, nozzles, high-speed air jets

Weber We	$\rho v^2 L/\sigma$	Inertia vs surface tension — droplets, spray dyeing, atomisation
Drag coefficient C_D	$F/(\frac{1}{2}\rho v^2 A)$	The Π_1 of the worked example, dressed with ½ and area

4. Similitude — making the model tell the truth

A model predicts its prototype when three similarities hold: **geometric** (same shape, scale ratio λ = L_p/L_m), **kinematic** (velocities scale consistently), and **dynamic** — the ruling dimensionless groups match. Complete dynamic similarity is usually impossible (matching Re *and* Fr with the same fluid demands contradictory velocities), so practice matches the group that governs the dominant physics and corrects or tolerates the rest.

WORKED EXAMPLES — TWO SIMILARITY LAWS

Reynolds similarity (pipe flow): a 0.3 m prototype pipe runs water at 3 m/s; a 0.1 m model uses the same water. Re equal: v_m × 0.1 = 3 × 0.3 → v_m = **9 m/s**. Small models of viscous flow must run *faster* — often impractically fast, which is why pressurised or different-fluid facilities exist.

Froude similarity (spillway, 1:25 model): velocity scale = √λ = 5. A prototype flood velocity of 10 m/s is modelled at **2 m/s**; time scales by 5 as well, and discharge by λ^(5/2) = 3125 — a 3125 m³/s prototype flood becomes a manageable 1 m³/s in the laboratory flume.

Textile engineering runs on this logic daily: pilot dyeing machines scale (or fail to scale) liquor ratio, flow and fabric speed to production machines; aerodynamic tests of opening rollers and ducts, and hydraulic models of effluent plants, all answer to Reynolds and Froude. Whenever a pilot result refuses to reproduce at scale, a mismatched dimensionless group is the first suspect.

5. Key points

- Homogeneity first: if the dimensions do not balance, the equation is wrong, whatever its source.
- Buckingham: n variables, m dimensions → n − m dimensionless groups; the sphere-drag list of five variables collapses to C_D = φ(Re).
- Match the group that governs: Re for viscous/pipe worlds, Fr where gravity and free surfaces rule, M when air gets fast.
- Scales: Froude velocity ∝ √λ, discharge ∝ λ^(5/2); Reynolds with the same fluid demands v ∝ 1/L.
- Perfect similarity is rare — engineering is choosing which group to honour and correcting the others knowingly.

6. Practice

Answer in your own words; brief model answers follow.

1. Check dimensionally whether the formula $h_f = f(L/D)(v^2/2g)$ is homogeneous, given f is dimensionless.
2. The period T of a simple pendulum is suspected to depend on length l, gravity g and bob mass m. Use dimensions to find the form of the law — and the surprise it contains.
3. A 1:10 model of a water pump is tested with the same water. Prototype speed 1450 rpm: what model speed gives dynamic (Reynolds/Froude-free, turbomachine) similarity of the flow pattern, using $v \propto ND$ scaling with $Re \propto ND^2/\nu$?
4. A ship model 1:50 is towed for wave resistance (Froude-governed). Prototype speed 8 m/s: find the towing speed.
5. Pressure drop Δp in a pipe is believed to depend on ρ, v, D and μ (ignoring length and roughness). Derive the Π groups and state the resulting law.
6. Why can a wind tunnel not match both Re and Mach number of a full-scale aircraft using the same air at the same temperature, unless it is pressurised or cooled?

Answer key

1. $v^2/2g$ has dimensions $(L^2T^{-2})/(LT^{-2}) = L$; L/D is dimensionless; so h_f has dimension L — head in metres ✓ homogeneous.
2. $T = K\, l^a g^b m^c$: $T^1 = L^a (LT^{-2})^b M^c \rightarrow c = 0$, $a + b = 0$, $-2b = 1 \rightarrow b = -½$, $a = ½$. $T = K\sqrt{(l/g)}$ — mass drops out entirely: the period is independent of the bob's mass, which dimensional analysis proves before any dynamics.
3. Re similarity with the same fluid: $N_m D_m^2 = N_p D_p^2 \rightarrow N_m = 1450 \times (10)^2 = 145{,}000$ rpm — absurd; real pump models accept distorted speed and correct by affinity laws and efficiency scale effects, a standard example of imperfect similarity managed knowingly.
4. Froude: $v_m = v_p/\sqrt{50} = 8/7.07 = 1.13$ m/s.
5. Variables Δp $[ML^{-1}T^{-2}]$, ρ, v, D, μ: $n = 5$, $m = 3 \rightarrow 2$ groups. With repeating ρ, v, D: $\Delta p/(\rho v^2)$ (Euler number) and $\rho vD/\mu$ (Re). Law: $\Delta p/(\rho v^2) = \varphi(Re)$ — pipe pressure coefficient is a function of Reynolds number (the seed of the Moody chart once L/D and roughness are added).
6. $Re \propto vL$ and $M \propto v$: shrinking L by 10 cuts Re tenfold at the same speed, but raising speed 10× to restore Re raises Mach tenfold too. Both can be restored only by changing the fluid state — raising density (pressurisation) raises Re without raising v, or lowering temperature raises density and lowers the speed of sound; hence pressurised/cryogenic tunnels.